

Summer Internship Assignment - 1

Submission Instructions

- Use Python 3.10 or above.
- Submit the assignment through a GitHub repository.
- Ensure all scripts run correctly.
- Avoid hardcoded paths.
- Use clean and maintainable code.
- Include output files and screenshots wherever necessary.
- Organize the project properly.
- Use modular and reusable code.
- Add meaningful comments and documentation.
- Create a proper README.md file & Include requirements.txt.

Task 1 — List & Dictionary Operations

- Take a list of integers as input.
- Remove duplicates and sort the list.
- Create a dictionary containing maximum value, minimum value, average value, and frequency of each element.
- Print the final output in a clean format.

Task 2 — File Handling

- Read a text file containing log entries as below: (if data not given generate random data)

INFO User logged in

ERROR Database timeout

WARNING Memory usage high

ERROR Invalid credentials

Write a script that :

- Count occurrences of each log type.
- Save the results into a JSON file.
- Print the most frequent log type.

Task 3 — Functions & Error Handling

- Create a function named `safe_divide(a, b)`.
- Handle division by zero.
- Handle invalid inputs.
- Return meaningful error messages.
- Add proper docstrings.

Task 4 — Debugging

- Fix the provided buggy Python code.

```
numbers = [1, 2, 3, 4, 5]
```

```
for i in range(len(numbers)):
    numbers.remove(numbers[i])
```

```
print(numbers)
```

- Explain why the bug occurs.
- Provide the corrected solution & Explain the time complexity of your approach.

Task 5 — Data Cleaning

- Load a public CSV dataset(<https://www.kaggle.com/datasets/aiexplorer77/digital-burnout-and-productivity-analytics>) using pandas.
- Handle missing values.
- Remove duplicate rows.
- Show dataset statistics.
- Encode categorical columns if required.
- Save the cleaned dataset.

Task 6 — Data Visualization

- Create at least five visualizations from the dataset.
- Use charts such as histogram, scatter plot, heatmap, bar chart, or boxplot.
- Add proper labels, titles, and legends.
- Save all plots as image files.

Task 7 — Insights Report

- Write a short report describing trends and observations from the dataset.
- Mention possible issues in the dataset.
- Summarize key conclusions.

Task 8 — Web Scraper

- Build a scraper using requests and BeautifulSoup.

Create a Python pipeline that:

1. Searches Twitter/X posts using keywords and context.
2. Extracts:
 - tweet text
 - username

- tweet link
 - video URL or media link (try to download video if not then just keep link/URL)
 - timestamp
 - engagement statistics (likes/reposts if available)
3. Stores the collected data in:
- CSV format
 - JSON format

Example search contexts:

- AI generated videos, Robotics demos, Self-driving cars, Sports highlights

Requirements:

- Use modular code
- Handle request failures and exceptions properly
- Add retry mechanisms where needed
- Maintain clean folder organization

Task 9 — Context Based Scraping

Extend the pipeline to:

1. Accept multiple keywords/topics from a file.
2. Collect videos from multiple searches.
3. Merge all collected results.
4. Remove duplicate tweets/videos.
5. Save videos into categorized folders based on context. Example:
 - videos/
 - |— robotics/
 - |— ai/
 - |— sports/
 - |— semiconductor/

Task 10 — Data Summary & Analysis

Generate: (at least 2 visualizations from the collected data below)

- total videos collected
- total unique creators
- most common keywords
- average engagement statistics
- top-performing posts
- Save Analysis into **summary_report.txt**